Hugo Marquerie 13/05/2025

Fórmula de D'Alembert

Teorema 1 (Fórmula de D'Alembert). *Sea* $u\colon (0,L)\times(0,\infty)\longrightarrow\mathbb{R}$ *dada por*

$$u(x,t)=\frac{1}{2}\left(f(x+t)+f(x-t)\right)+\frac{1}{2}\int_{x-t}^{x+t} g(s)\,\mathrm{d}s \quad \textit{donde} \quad f,g\colon\mathbb{R}\to\mathbb{R}.$$

Entonces, u *es solución de la ecuación de ondas en dimensión* 1 *homogénea*

$$\forall x\in(0,L), t>0 : u_{tt}-u_{xx}=0 \text{ con contorno Dirichlet } \forall t>0 : u(0,t)=0=u(L,t)$$

y condiciones iniciales $\forall x\in(0,L) : u(x,0)=f(x) \wedge u_t(x,0)=g(x)$ *con* f,g *conocidas.*

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